Ysbyty Gwynedd
Anaesthetics Department
Adran Anaestheteg

PRE-OPERATIVE ASSESSMENT & PREPARATION

Obstructive Sleep Apnoea

YG CG 310 Mar23

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FIRST APPROVED: JULY 2019 REVIEW BY: JULY 2025

Key Points

- Patients with obstructive sleep apnoea (OSA) are at an increased risk of perioperative complications.
- Any pre-existing continuous positive airway pressure therapy should be continued in the post-operative period.

Introduction

- Obstructive sleep apnoea (OSA) affects 5–10% of the adult population with estimates of twice that in patients scheduled for surgery.
- Up to 85% are undiagnosed, therefore untreated.
- Patients with OSA are at increased risk of stroke, glucose intolerance, systemic and pulmonary hypertension, myocardial infarction & right heart failure.
- OSA is also associated with difficult intubation.
- OSA can be worse following sedatives / anaesthesia / opioids.

The severity of OSA is determined by the apnoea/hyponoea Index (AHI) which is obtained by conducting formal sleep studies:

- AHI 0-5 = “normal”
- AHI 5-15 “mild” OSA
- AHI 15-30 “moderate” OSA
- AHI >30 “severe” OSA

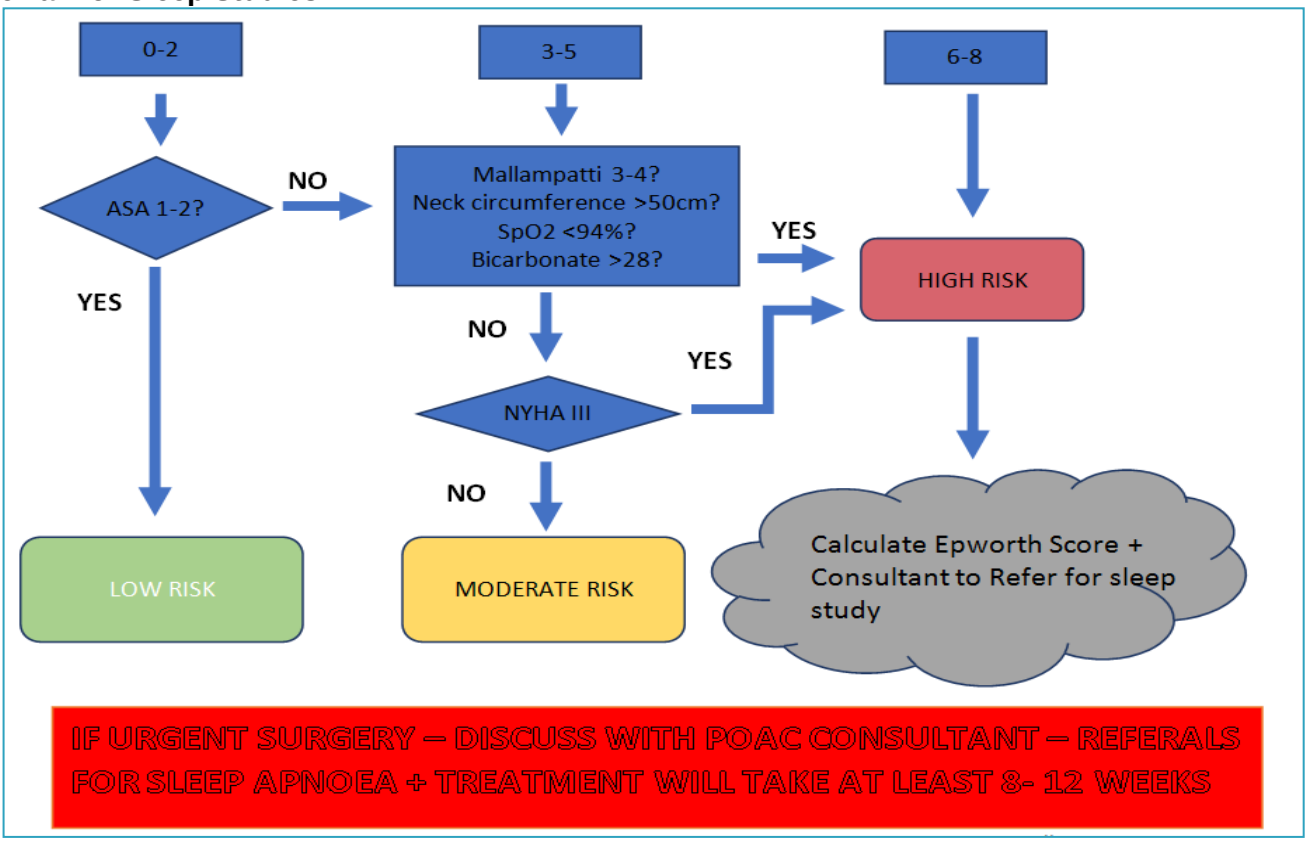
Pre-Operative Screening

- The validated tool for screening for sleep apnoea is STOP-BANG, which is superior to the Epworth Score. Score one point for each positive answer.
- All patients with a BMI > 35 should be screened.
- N.B. Patients who do not snore are unlikely to have OSA

Snoring	Do you snore loudly (louder than talking or heard through a closed door?)
Tired	Do you often feel tired, fatigued or sleepy during the daytime? Do you fall asleep in the daytime?
Observed	Has anyone observed you stop breathing or choking or gasping during your sleep?
Blood **P**ressure	Do you have, or are you being treated for, high blood pressure?
BMI	BMI > 35 kg.m^2
Age	>50 years
Neck	Is your Neck circumference (measured around Adam's apple) > 43 cm (17 in)? (MALE) or > 41 cm (16 in)? (FEMALE)
Gender	Male gender?

- If the STOP- BANG score is ≥3, then request **bicarbonate** levels in addition to any standard blood tests.
- If your patient is 'high risk', then in addition to STOP-BANG, complete the **Epworth score questionnaire** with the patient so this can be included in the referral letter.

Referral For Sleep Studies:



- High risk patients are referred using an agreed standard referral template.
- Ideally, 8 to 12 weeks of pre-operative treatment with CPAP is required.

Urgent or cancer surgery

If surgery cannot be delayed for establishment of CPAP therapy, then, where there is a strong index of suspicion on clinical grounds or on using the screening tool, patients should be managed peri-operatively as if they have sleep apnoea.

Elective booking of HDU

POAC will book HDU bed for patients:

1. With known moderate & severe OSA
2. all patients identified in POAC as high risk for OSA (ie referred to Respiratory)

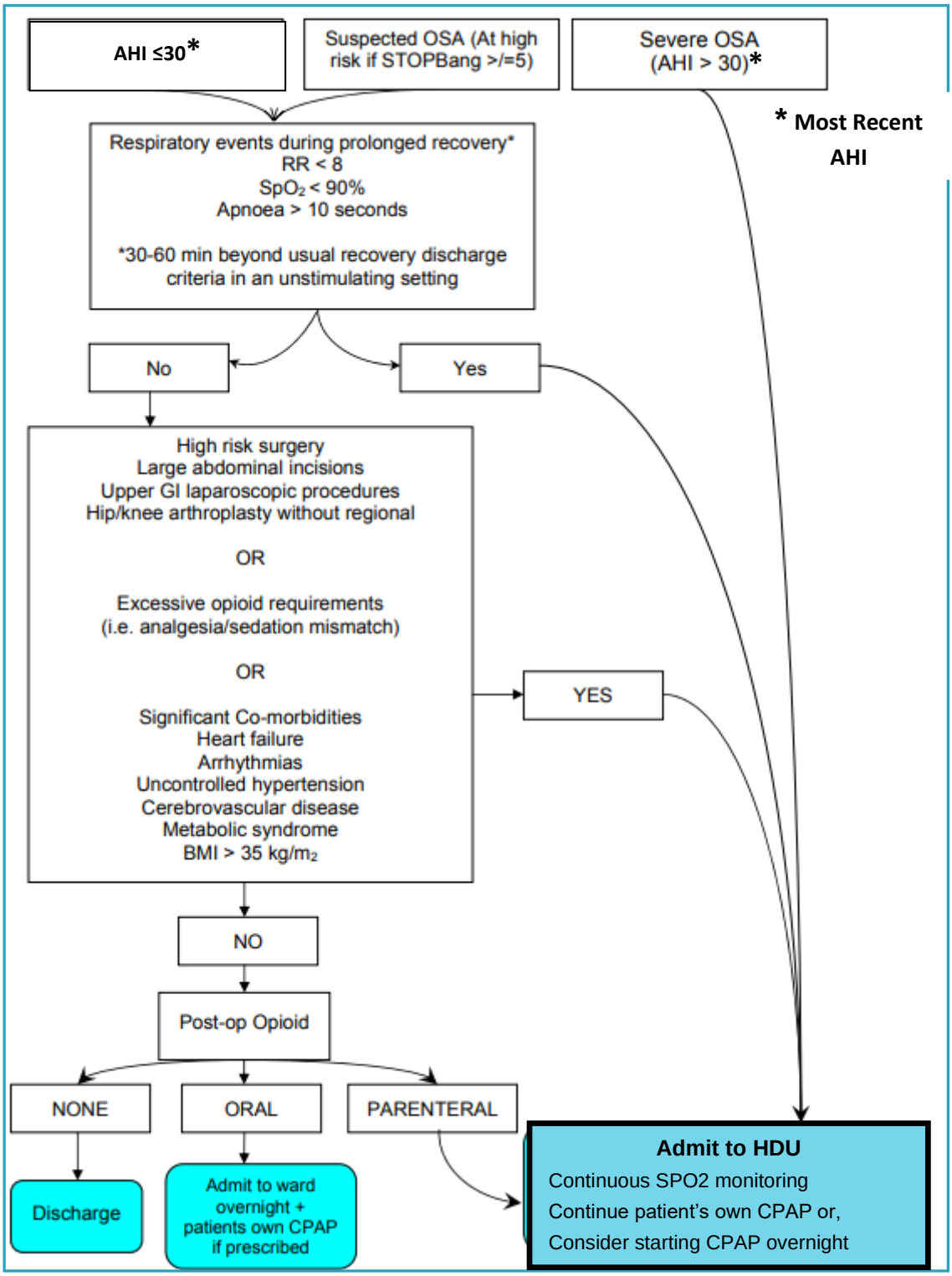
Peri-Operative Management

- If patient uses CPAP at home ensure has own CPAP machine with them (can you work it?)
- Avoid sedative pre-medication.
- Avoid General Anaesthesia and sedation where possible.
- If Severe OSA, consider arterial line for ABG monitoring.
- Anaesthetise in ramped up position.
- Pre-oxygenate with PEEP
- Use short acting anaesthetic agents and use opioid sparing techniques.
- Ensure full muscle relaxant reversal utilising objective NMT monitoring.

Management in Recovery

- Extended Recovery for **all grades** of OSA (mild-severe or suspected)
    - ie. Observe for 30-60 minutes beyond time usual discharge criteria met.
- Nurse upright and mobilise early
- Observe for the following whilst unstimulated:
    - Apnoeas (>10 seconds)
    - SpO2 <90%
    - Resp rate <8 breaths per min
- If any occur, then patient requires HDU for continuous post-operative monitoring overnight.
- Patients with Mild OSA (AHI<15) can otherwise be managed as any other non-OSA patient.

Post-Operative Management



N.B. Remember to cancel any pre-booked HDU bed, if no longer required.

References

- OSA. CEACCP 2011, Vol 11,1,:5-8.
- The Society for Obesity and Bariatric Anaesthesia OSA Guidelines, Feb 2016
- Royal Cornwall Hospitals NHS Trust CG for Per-Operative Management of Adults with Obstructive Sleep Apnoea, Jan 2016.
- Sheffield Teaching Hospitals NHS Trust pre-Operative Guidelines, March 2017

Appendix 1: Referral Letter

GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board

Pre-Operative Assessment Clinic

Telephone: 01248 384177 / 384366 / Fax: 01248 384920

Ref: SA/XX/DXXXXXX
Date:
Clinic Date:

Dr D McKeon
Consultant Respiratory Medicine, Sleep Clinic
Ysbyty Gwynedd
Bangor
LL57 2PW

Dear Dr McKeon,

Patient Addressograph

Please could you review this patient in the sleep clinic who has been assessed as high risk for possible obstructive sleep apnoea as soon as is convenient?

Patient History

- PMH:
- Occupation:
- Current Medications:

Sleep Apnoea details

- Epworth score:
- Symptoms and signs:

This patient is listed for surgery. The planned operation is:_____

Unless urgent, the planned surgery will be deferred until after your review. If the patient requires CPAP therapy, surgery will be deferred until this is established in order to reduce the risk of critical care admission. We would be grateful therefore if you could facilitate this in order to minimise delays.

Many Thanks,

_____,
Consultant Anaesthetist